

SOURAV RAJ

Patna, Bihar

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PROFESSIONAL SUMMARY

Experience: Fresher

Data Scientist with strong foundations in **machine learning, statistics, and NLP**. Experienced in building **predictive models, scalable ML pipelines, and cloud-deployed applications** using Python, SQL, and AWS.

PROJECTS

Real Estate Price Prediction System (ML & Dashboard)

Oct 2025 – Dec 2025

- Web-scraped and aggregated **40K+ real estate listings from MagicBricks (for educational purposes)**; performed extensive **data cleaning, missing value imputation, and feature engineering** to prepare a high-quality modeling dataset.
- Engineered **13 predictive features** improving model robustness and trained a **LightGBM regression model achieving $R^2 = 0.90$ and MAE ≈ 0.6 Cr INR** using Optuna-driven hyperparameter tuning.
- Built an end-to-end **ML pipeline using MLflow, DVC, Docker, and AWS (S3, EC2, ECR)** with **GitHub Actions CI/CD** for automated experiment tracking, versioning, and deployment.
- Deployed a **Streamlit web application** enabling real-time property price prediction and interactive dashboards for market insights.

Twitter Sentiment Analysis (NLP & Deep Learning)

Aug 2025 – Present

- Performed EDA on **69K+ tweets** to analyze class imbalance, text distribution, and noise patterns for model design decisions.
- Developed a **PyTorch-based LSTM model achieving 88% accuracy, 88% macro F1-score, and 0.98 AUC** on the test dataset.
- Integrated **MLflow with Optuna** for experiment tracking and tuning.
- Deployed inference application on **Streamlit Cloud** enabling real-time sentiment predictions.

Hybrid Spotify Recommender System

Apr 2025 – May 2025

- Designed hybrid recommender combining **collaborative and content-based filtering on 1M+ user records and 50K+ music metadata entries**.
- Engineered similarity-based user and item representations using **cosine similarity**.
- Implemented scalable recommendation pipeline using **Docker, DVC, GitHub Actions (CI/CD), and AWS (S3, EC2, ECR, CodeDeploy)**.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, SciPy), SQL (MySQL)

Machine Learning: Scikit-learn, XGBoost, LightGBM, Regression, Classification, Clustering, Model Evaluation

Deep Learning: PyTorch, TensorFlow, Keras

Statistics: Hypothesis Testing, Feature Engineering, A/B Testing

MLOps & Deployment: MLflow, DVC, Docker, FastAPI

Cloud: AWS (EC2, S3, ECR, CodeDeploy)

Visualization: Matplotlib, Seaborn, Power BI, Excel

Version Control: Git, GitHub, GitHub Actions

Generative AI: LangChain, LangGraph, Retrieval-Augmented Generation (RAG)

EDUCATION

Bachelor of Science in Mathematics: Scored (76%)

2020 – 2023

Patna Science College, Patna

CERTIFICATIONS

Data Science Mentorship Program 2.0 – CampusX

Jan 2026

Machine Learning A-Z: AI, Python & R – Udemy

July 2026